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COMPARATIVE ANALYSIS OF MACHINE LEARNING MODELS FOR USER VERIFICATION VIA KEYSTROKE DYNAMICS

A report on
Machine Learning Project
[CSE-4408]

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COMPARATIVE ANALYSIS OF MACHINE LEARNING MODELS FOR USER VERIFICATION VIA KEYSTROKE DYNAMICS

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Abstract - This study presents a comprehensive and comparative analysis of four different machine learning models for user authentication using keystroke dynamics. The models evaluated include SVM, Random Forest , Logistic Regression and XGBoost. Using the CMU Keystroke Dynamics Benchmark Dataset consisting of 20400 rows ,we compare the approaches between classification and regression tasks. Performance was majorly gauged through ROC-AUC curve and confusion matrix. Our findings show that XGBoost and Random Forest outperform the other models in classifying impostor and genuine user.

I. Introduction

Everyone uses a computer in their daily routine life. Many applications or websites authenticate the users by giving them usernames and emails along with the password. But passwords can get compromised and hence lead to issues with security and privacy. In order to tackle this problem, many other types of authentication have emerged, including face recognition and fingerprint recognition. Though this provides security to users, these authentication techniques tend to be costly and need specific hardware devices. The keystroke dynamics technique is one more way of identification, which makes use of the unique typing styles of the user such as how long a person takes in pressing a particular key and the interval between pressing two consecutive keys. Even when the user uses the correct password, it helps in recognizing whether it is the right individual trying to login. This system does not require the use of any expensive hardware.

II. Literature review

A. Keystroke Dynamics: Concepts, Techniques, and Applications:

The current research evaluates the usage of machine learning models for smart user authentication using keystroke dynamics. In particular, this research involves a novel dataset and machine learning models such as KNN, Random Forest, and Gradient Boosting. This research comprises three major stages, including data preprocessing, feature extraction, and evaluation of machine learning models. The findings reveal the effectiveness of the application of machine learning models for the discrimination of users based on the analysis of their keystrokes dynamics. The outcomes indicate the importance of the selection of appropriate features for the enhancement of machine learning models. The findings of the paper highlight the possibility of enhancing security mechanisms through the continuous authentication process. Additionally,

the paper highlights the effectiveness of the proposed approach since it enables comparing the efficiency of various machine learning algorithms.

B. Keystroke Dynamics Analysis Using Machine Learning Methods:

In this study, the proposed model is concerned with employing various machine learning techniques for user identification based on keystroke pattern recognition. Our analysis treats the identification issue as a classification problem and employs algorithms such as Logistic Regression, Support Vector Machines, Random Forests, and Neural Networks to analyze it. The data used in the dataset consists of information related to the timing of keystrokes; normalization techniques are then performed on the data before applying the algorithms to train them. The experiments demonstrate that all algorithms are highly effective in terms of accuracy, but the neural network performs better than the rest.

C. A Systematic Literature Review on Latest Keystroke Dynamics Based Models:

The methodology used in this paper is the comprehensive analysis of the latest advancements in keystroke dynamics from 2017 till 2022. In doing so, various articles are discussed and classified based on their approaches to authentication, identification, and predictions. Furthermore, there is mention of issues associated with the traditional password-based authentication system and how keystroke dynamics can address security concerns by providing solutions to those issues. Various issues include data variability issues, feature extraction techniques, and practicality concerns. This paper finds that the area of keystroke dynamics has great potential in the field of cybersecurity.

D. Enhancing Keystroke Dynamics Authentication with Ensemble Learning:

This research paper by Xiaofei Wang and Daqing Hou focuses on improving the performance of the keystroke authentication system using the concept of ensemble learning. In the study, issues such as the imbalance of data and the poor performance of conventional classifiers are handled. To overcome these challenges, resampling approaches and classifier methods such as Random Forest and XGBoost have been implemented in the study for accuracy purposes. It is evident from the findings that ensemble learning approach is far much better compared to the conventional method used since it yields higher levels of accuracy. Furthermore, the use of multiple models enables identification of complex typing patterns.

E. User Attribution in Digital Forensics Using Keystroke and Mouse Data:

In this paper, the use of keystroke dynamics in the field of digital forensics is examined with an objective to identify users based on their behavior. Machine learning algorithms such as support vector machine (SVM) and eXtreme Gradient Boosting (XGBoost) have been utilized to analyze the combined information of keystrokes and mouse clicks. The importance of behavioral biometrics in the process of tracking user identity in cyber forensics is emphasized in this paper. In addition, various techniques based on statistics and machine learning can be utilized for classification purposes. Moreover, the fusion of multiple behavioral metrics increases the reliability of the proposed system.

F. Keystroke Dynamics for Intelligent Biometric Authentication:

In this study, the researchers examine how machine learning models can be applied to perform smart authentication based on keystroke dynamics. This study employs the novel dataset and uses various algorithms including KNN, Random Forest, and Gradient Boosting. The experiment includes the following stages: data preprocessing, feature extraction, and modeling. It turns out that the machine learning models have proven to work well in recognizing typing behaviors by means of appropriate evaluation metrics. Besides, the importance of selecting the proper features in order to increase model efficiency is

demonstrated. Finally, this study proves that the keystroke dynamics approach can be useful for enhancing the existing security systems and making them reliable and convenient. Furthermore, the comparison of different algorithm performances becomes easy due to identical experimental conditions.

III. METHODOLOGY

The project focuses on analyzing and comparing different machine learning models to verify users based on their typing patterns. The entire process is divided into three main stages: data collection and feature extraction, data preprocessing and model training with tuning.

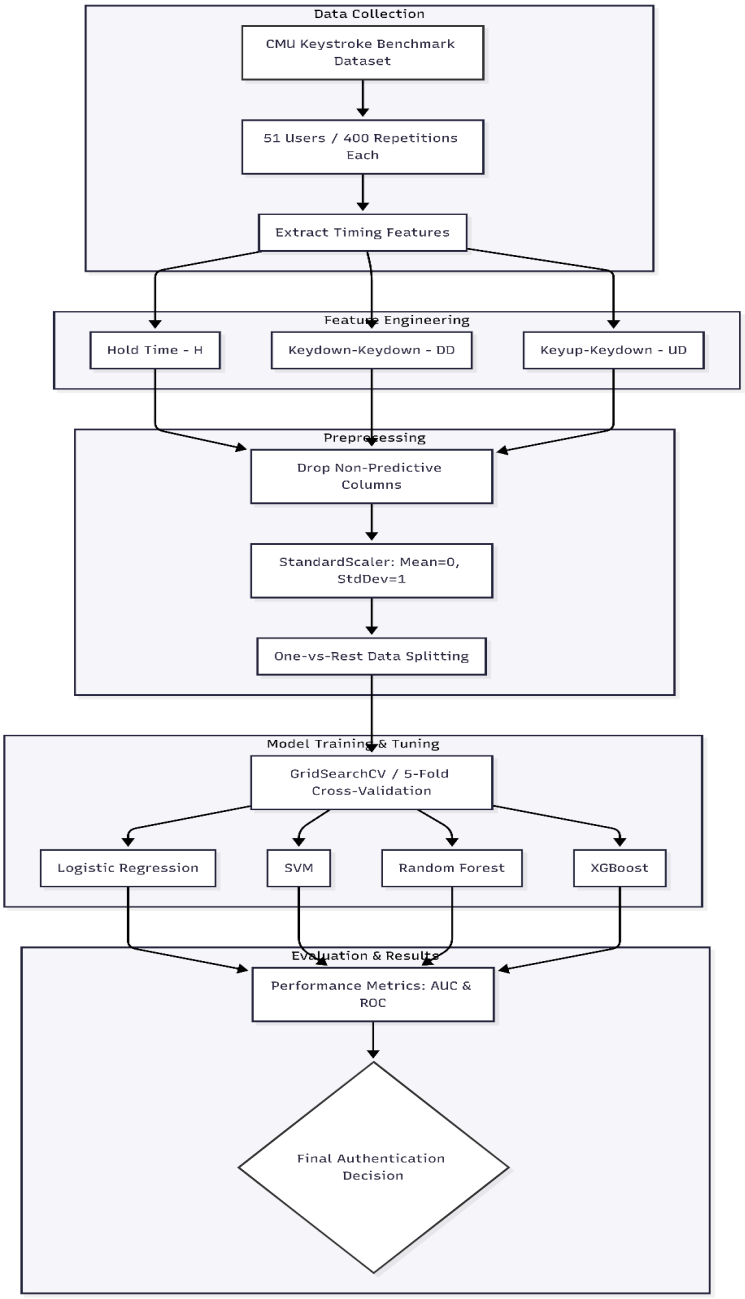

Fig 1. Machine Learning Workflow for Keystroke Dynamics Authentication

A. *Data collection and feature extraction*

The dataset that was employed in the present work is the CMU Keystroke Dynamics Benchmark Dataset. It includes typing information obtained from 51 participants, wherein each participant types a fixed password ".tie5Roanl". For every participant, there are 400 observations (8 sessions * 50 repetitions).

Timing-based features are derived from each typing observation. They include:

- Hold Time (H): The time difference between pressing and releasing the key
- Keydown-Keydown (DD): The time difference between pressing two consecutive keys
- Keyup-Keydown (UD): The time difference between releasing one key and pressing the next
-

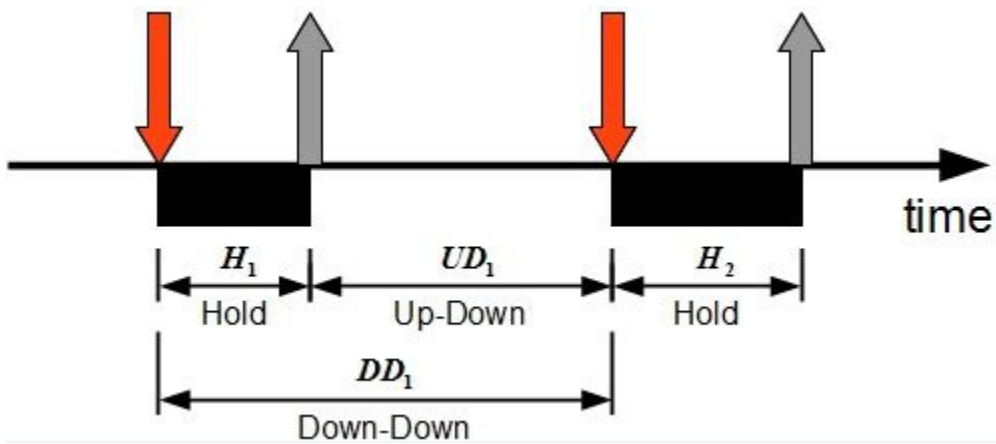

Fig 2. Visualization of the key features in keystroke dynamics

In total, 31 such features were used to represent each typing sample. These features together form a unique typing profile for every user.

B. *Data Preprocessing*

Before feeding the data into the models, a few important steps were performed to clean and prepare it.

- *Feature Selection:* Columns like sessionIndex and repetition count were dropped since they do not help in identifying users.
- *Standardization***:** Since the Hold time will be lesser than DD time and UD time, it is important to ensure that Hold time doesn't get suppressed by the features with higher values. So, StandardScaler was used to normalize them. This ensures all features are treated equally by the model.
  The formula used is:
  $$z = \frac{x - \mu}{\sigma}$$
  where (μ) is the mean and (σ) is the standard deviation.
  This step adjusts all values so that they are centered around 0 and have similar spread.
- *Data Splitting (One-vs-Rest):*
  To simulate real-world authentication, a one-vs-rest approach was used.
  For each user:
  - Genuine samples were taken from that user
  - Impostor samples were taken from other users

This helps the model learn to distinguish between the actual user and others trying to mimic them.

c. *Model implementation and hyperparameter tuning*

The following four types of machine learning algorithms were used and compared:

- Logistic Regression: Simple linear algorithm used as the baseline model.
- Support Vector Machine (SVM): Algorithm that identifies an optimal boundary between the actual users and fraudsters.
- Random Forest: Bagging algorithm that constructs decision trees en masse.
- XGBoost: Boosting algorithm that constructs decision trees iteratively.

To maximize the efficiency of the selected algorithms, hyperparameter tuning was performed via GridSearchCV using a 5-fold validation scheme.

Some of the key hyper-parameters that have been selected are:

- *C (Support Vector Machines)*: It determines the complexity of the algorithm. The higher its value, the more accurate the classification, whereas the lower its value, the more error tolerant it will be.
- *Kernel (Support Vector Machines)*: The type of decision boundary (whether linear or non-linear).
- *n_estimators (Random Forest)*: It determines the number of trees in the forest.
- *Learning Rate (Extreme Gradient Boosting)*: Determines how influential each new tree will be.
- *Max Depth (Extreme Gradient Boosting)*: Limits tree depth.

IV. IMPLEMENTATION DETAILS

All the models were trained and tested using scikit-learn along with other machine learning packages. In the beginning, data pre-processing and splitting was done based on the one-vs-rest scheme. Thereafter, all the models were trained and tested on their respective training and testing datasets.

In keystroke verification, acceptance of authentic users and rejection of impersonators have similar levels of significance. That is why the ROC-AUC score has been chosen as the primary evaluation criterion since it takes into account the tradeoff between False Acceptance Rate (FAR) and False Rejection Rate (FRR).

False Acceptance Rate (FAR):

FAR = (Number of Impostor Attempts Accepted) / (Total Number of Impostor Attempts)

False Rejection Rate (FRR):

FRR = (Number of Genuine Attempts Rejected) / (Total Number of Genuine Attempts)

Accuracy and related evaluation metrics such as precision and recall only evaluate model performance at a particular decision threshold whereas ROC-AUC provides a much more holistic overview of model performance at all possible decision thresholds.

V. RESULTS AND DISCUSSIONS

The AUC values for Random Forest and XGBoost were almost equal (~0.995), implying that these models can easily distinguish between legitimate and fake users. Between the two models, Random Forest showed better performance since it could accurately identify impostors, i.e., the number of false acceptances was low. The ROC curves for both models are close to the upper left corner of the graph, signifying that both models have low FAR values (~2-3%). This indicates that the systems are less prone to false acceptances, thereby providing a safe environment for legitimate users.

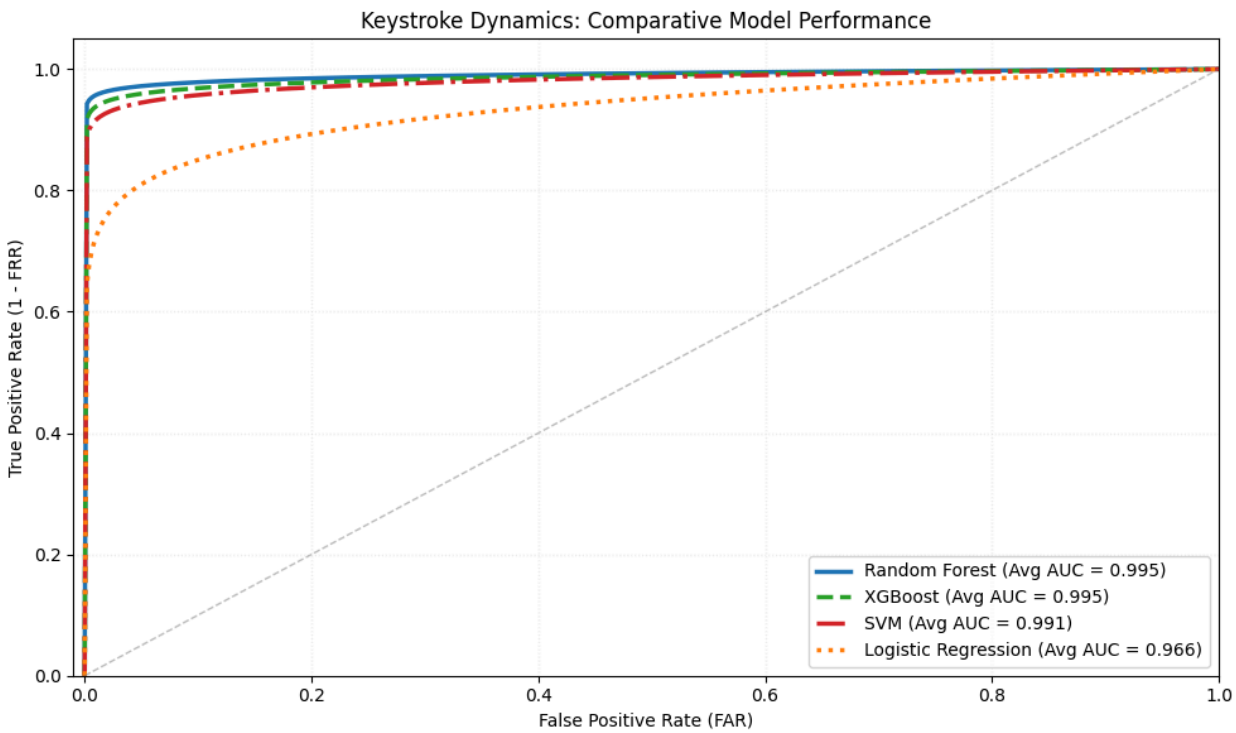

Fig 3. ROC Curve analysis of the four implemented models.

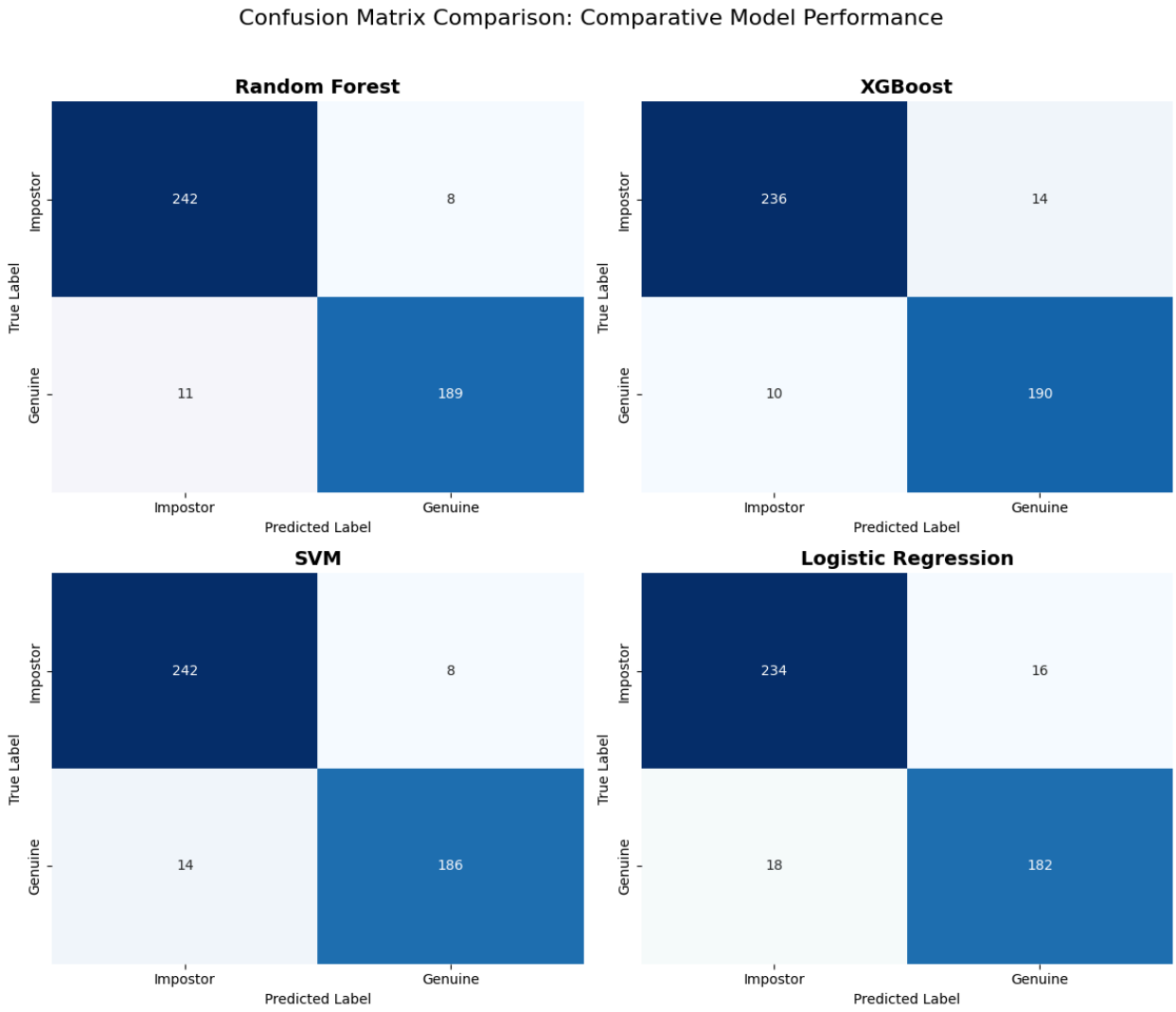

Fig 4. Confusion matrix analysis of the four implemented models.

VI. CONCLUSION

In the current dataset, all four models demonstrated a great level of AUC (more than 0.96). This means that individual typing rhythms can be effectively utilized for two-factor authentication purposes. The Random Forest and XGBoost models outperformed the Logistic Regression model, demonstrating a very close level of AUC (0.995). It is possible to conclude that the data created based on extracted features cannot be separated linearly. Therefore, the logistic regression algorithm underfits and fails to account for the complex pattern of typing rhythms. Keystroke dynamics appears to be one of the promising directions for the future with its capability to deliver efficient and affordable methods for two-factor authentication during a time when password hacks, data breaches, and online frauds become prevalent.

VII. FUTUREWORK

One possible way to enhance the proposed method is by evaluating it using large-scale datasets in order to validate that it would work effectively regardless of the user's typing style. Moreover, additional features about the typing process can also be considered to further enhance the accuracy of the results. On the other hand, another approach would be to utilize deep learning algorithms that would allow the system to model complex behavior.

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CONTRIBUTIONS:

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Aditi Gupta	XGBoost
Biash Gazi	SVM
Bhavya Singhvi	Logistic regression